School of Engineering and Technology, MGM University Chh. Sambhajinagar

Academic Year 2025-2026

Mini Project report
On

"Banking Sector "

Submitted in partial fulfillment for Eighth Semester of B.Tech (Integrated) in

COMPUTER ENGINEERING

(Data Science)

By

Mohd Asif Khan	**202256112040**
Mohd Mudabbir Mohd Sadique Nadvi	**202256112045**
Amaan Maniyar	**202356112701**
Zaid Khan	**202356112703**

Under the guidance of
Ms. Rathi S.R.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This to certify that **Mohd Asif Khan, Mohd Mudabbir Mohd Sadique Nadvi, Amaan Maniyar, Zaid Khan** with PRN No **202256112040, 202256112045, 202356112701, 202356112703** has successfully completed their Mini-Project entitled "**Banking Sector"** in the Course of **"Data Analytics"** in the Eighth Semester during tenure of completing the B. Tech(I) program in **Computer Science and Engineering (Data Science** from **School of Engineering and Technology, MGM University** institute.

Ms. Rathi S.R.	**Ms. Rathi S. R.**
Guide	**HOD**
CSE Departmen	CSE Department

CONTENT

Sr. No.	Title	Page No.
1	Abstract	04
2	Introduction	05
3	Problem Statement	07
4	Course Outcomes Integrated	09
5	Action Plan	10
6	Literature Review	11
7	Industry Overview	13
8	Data Description & Preparation	16
9	Proposed Methodology	18
10	Graphical Representations & Analysis	20
11	Statistical Measures	22
12	Implementation & Output	24
13	Interpretation & Business Insights	25
14	Applications of this Project	27
15	Skills Developed	28
16	Area of Future Improvement	29
17	Conclusion	30
18	References	32

1. ABSTRACT

This mini project report presents a comprehensive descriptive analysis of the Indian Banking Sector spanning eleven years from 2013 to 2023. The study focuses on four key financial variables: Deposits, Credit (Loans and Advances), Non-Performing Assets (NPAs), and Net Profits. Using statistical tools and graphical representations, this report examines long-term trends, measures of central tendency and variability, distributional patterns, and inter-variable correlations.The analysis reveals that while the banking sector demonstrated consistent growth in deposits and credit over the decade, it faced a severe crisis between 2016 and 2018 due to the escalation of Non-Performing Assets (NPAs). This period saw significant declines in profitability and credit growth. However, the sector showed remarkable resilience through regulatory interventions,digital transformation, and improved risk management practices, leading to a strong recovery post-2019.

The project dashboard (BankSight Pro) was developed as an interactive analytics tool covering all three banking sectors Public, Private, and Foreign Banks with comprehensive chart analysis including Line Charts, Bar Charts, Pie Charts, Scatter Plots, and Area Charts. Key Performance Indicators tracked include Total Deposits (₹1,75,000 Cr in 2023), Total Credit (₹1,58,000 Cr), NPA Level (2.7%), and Net Profit (₹16,800 Cr).The findings underscore the critical relationship between asset quality and profitability, the importance of data-driven decision-making in financial management, and the transformative role of digital banking in reshaping the sector. This report provides data-driven business insights for strategic planning, risk mitigation, and future growth of the Indian banking industry.

2. INTRODUCTION

2.1 Overview of the Banking Sector

The banking sector is one of the most essential and influential pillars of any national economy. It serves as the backbone of financial development and plays a critical role in supporting economic growth, stabilizing financial systems, enabling credit availability, and promoting savings mobilization. The sector facilitates smooth financial transactions, provides secure deposit services, extends credit to businesses and individuals, and helps maintain liquidity in the economy.

Banks act as intermediaries between surplus and deficit economic units, thereby enabling optimal resource allocation. Savings collected from depositors are transformed into productive investments through various loan products, which contribute to industrial expansion, entrepreneurship, employment generation, and overall economic stability. The banking system also supports government policies related to monetary control, inflation management, interest rate determination, and financial inclusion.

2.2 The Indian Banking Landscape

In India, the banking industry is regulated by the Reserve Bank of India (RBI), which ensures financial stability, oversees monetary policy, regulates interest rates, and protects the integrity of the financial system. Over the past decade, the Indian banking sector has undergone profound transformation due to technological disruption, regulatory reforms, digital finance growth, and the rise of fintech participants.The introduction of Unified Payments Interface (UPI), Aadhaar-based KYC, instant mobile banking, and online payments has revolutionized customer experiences and dramatically increased the accessibility of banking services. As of 2023, India processes billions of UPI transactions monthly, a testament to the sector's technological leap forward.

2.3 Significance of Data Analytics in Banking

Data analytics has become indispensable in the banking environment. It supports risk assessment, customer segmentation, fraud prevention, credit scoring, loan recovery, and operational efficiency. Banks that leverage data effectively are better positioned to manage non-performing assets, optimize their loan portfolios, detect fraud early, and deliver personalized financial products.

2.4 Objectives of the Study

- To analyze historical trends in deposits, credit, NPAs, and profits over 2013–2023 across Public, Private, and Foreign Banks
- To measure central tendency and variability of key banking variables
- To identify correlations between risk indicators (NPA) and performance metrics (Profit, Credit)
- To detect unusual patterns, outliers, and crisis-period anomalies
- To build an interactive dashboard (BankSight Pro) for visual exploration of banking data
- To derive strategic recommendations based on data-driven analysis
- To understand the impact of regulatory reforms and digital transformation on banking performance

3. PROBLEM STATEMENT

"To analyze the historical financial performance of the Indian banking sector across Public, Private, and Foreign Banks using Deposits, Credit, NPAs, and Profits over the period 2013–2023, with the objective of identifying long-term trends, evaluating risk and volatility, interpreting statistical patterns, examining variable relationships, and generating data-driven business insights through an interactive analytics dashboard.".This problem is descriptive, analytical, and interpretative in nature. It does not require the development of a predictive model but instead focuses on understanding what happened, how the sector evolved, and what patterns emerge from 11 years of structured financial data covering three distinct banking segments.

3.1 Research Questions

- How did the Indian banking sector perform over the last decade (2013–2023)?
- How do Public Sector, Private Sector, and Foreign Banks differ in performance?
- What is the relationship between credit expansion and deposit mobilization?

3.2 Scope of the Study

The study covers the aggregate and sector-wise banking data of India for the period 2013 to 2023, involving four primary variables: Deposits, Credit, NPA (%), and Net Profit across three banking sectors — Public Sector Banks, Private Sector Banks, and Foreign Banks. The BankSight Pro dashboard enables multi-sector comparison, year-by-year analysis, comprehensive charting, and data management. The study employs descriptive statistical methods, graphical analysis, and correlation assessment.

3.3 Analytical Goals

Central Questions of Analysis

- How did the Indian banking sector perform over the last decade across all three segments?
- What insights can be drawn using descriptive data analytics and interactive visualization?
- What are the risk-return dynamics and sector-wise divergences in the Indian banking?
- How did digital transformation change the banking landscape post-2019?
- What strategic actions should banks take based on data-driven evidence?

4. COURSE OUTCOMES INTEGRATED

This mini project integrates several key course outcomes from the related academic curriculum. By working on this banking sector analysis and building the BankSight Pro interactive dashboard, students develop competencies in data handling, statistical reasoning, visualization, and business interpretation.

CO No.	Course Outcome Description	Integration in Project
CO1	Apply descriptive statistical methods to real-world datasets	Statistical measures — mean, median, SD, skewness applied to banking data across 3 sectors
CO2	Visualize and interpret data using appropriate graphical tools	Line, Bar, Pie, Scatter, Area charts implemented in BankSight Pro dashboard
CO3	Analyze business problems	NPA-profit relationship analyzed for strategic

CO No.	Course Outcome Description	Integration in Project
CO4	Understand data cleaning and transformation techniques	Missing value handling, outlier treatment, normalization techniques applied
CO5	Draw meaningful conclusions from quantitative analysis	Business insights derived from trend analysis and correlation studies

5. ACTION PLAN

The following action plan outlines the structured approach taken to complete this mini project. Each activity was assigned to team members and completed within defined timelines.

S.No.	Details of Activity	Planned Start	Planned Finish	Responsible Members
1	Topic selection and problem framing	Week 1	Week 1	All Members
2	Data collection and dataset compilation (2013–2023)	Week 2	Week 2	ZAID KHAN, AMAAN MANIYAR
3	Data cleaning, validation, and transformation	Week 3	Week 3	AMAAN MANIYAR, MOHD ASIF KHAN
4	Descriptive statistical analysis (mean, SD, skewness)	Week 4	Week 5	ZAID KHAN, MOHD ASIF KHAN
5	BankSight Pro dashboard development	Week 5	Week 6	AMAAN MANIYAR, MOHD MUDABBIR
6	Graphical representations (Line, Bar, Pie, Scatter, Area)	Week 6	Week 7	All Members

S.No.	Details of Activity	Planned Start	Planned Finish	Responsible Members
7	Business insights and correlation analysis	Week 7	Week 8	ZAID KHAN, AMAAN MANIYAR
8	Report writing and documentation	Week 8	Week 9	All Members
9	Review, formatting, and finalization of report	Week 9	Week 10	All Members
10	Presentation preparation and final submission	Week 10	Week 10	All Members

6. LITERATURE REVIEW

6.1 Evolution of Banking Sector Research

The study of banking sector performance has been a central topic in financial economics for decades. Research in this domain has evolved from simple balance sheet analysis to complex multi-variable studies incorporating risk assessment, regulatory impact, and technological transformation. The Indian banking sector has attracted significant scholarly attention due to its unique challenges — serving a large, diverse population while managing credit risk, regulatory compliance, and rapid digital adoption.

6.2 Non-Performing Assets — Literature Insights

Non-Performing Assets (NPAs) represent one of the most critical risk indicators in banking. Extensive literature confirms that rising NPAs negatively impact profitability, restrict credit growth, and erode capital adequacy. Studies by researchers at the RBI and academic institutions consistently highlight that the NPA crisis in India (2016–2018) was precipitated by overleveraged corporate borrowers, weak underwriting standards, and insufficient credit monitoring systems.

6.3 Credit-Deposit Dynamics

The relationship between credit and deposits has been extensively studied as an indicator of banking sector health. A high Credit-Deposit (CD) ratio indicates aggressive lending, while a very low ratio suggests under-utilization of deposits. Indian banking literature suggests an optimal CD ratio between 70–80%, balancing liquidity with productive deployment of funds. Research demonstrates that during periods of economic expansion, credit grows faster than deposits as businesses borrow more.

6.4 Digital Transformation in Banking

The digitization of banking has been a dominant theme in recent literature. The introduction of UPI in 2016, followed by Aadhaar-based KYC and digital lending platforms, fundamentally transformed how Indians interact with financial institutions. Studies indicate that digital banking reduces operational costs by 40–60% compared to branch-based services. Fintech research highlights that AI-powered credit scoring, fraud detection algorithms, and customer segmentation tools enable faster, more accurate lending decisions while significantly reducing default rates.

6.5 Sector-wise Performance Divergence

Literature extensively covers differences in performance among Public Sector Banks (PSBs), Private Sector Banks, and Foreign Banks. Foreign banks, despite having smaller balance sheets, consistently demonstrate superior NPA management — as evidenced in our dashboard showing Foreign NPA with only +5.56% change from 2013–2023 versus Total NPA at +17.39%. Private banks have demonstrated greater agility in digital adoption, while PSBs bear greater responsibility for financial inclusion and government-directed lending schemes.

6.6 Regulatory Frameworks and Their Impact

Literature extensively covers the regulatory interventions that reshaped Indian banking post-2015. The introduction of the Insolvency and Bankruptcy Code (IBC) in 2016, the Prompt Corrective Action (PCA) framework, and the consolidation of Public Sector Banks (PSBs) are all cited as critical turning points. These measures improved balance sheet transparency, forced recognition of hidden NPAs, and ultimately laid the foundation for sector recovery after 2018.

7. INDUSTRY OVERVIEW

7.1 Structure of the Indian Banking System

The Indian banking system is one of the largest and most complex financial systems in the world. It operates under the regulatory oversight of the Reserve Bank of India (RBI) and consists of multiple institutional categories, each serving distinct economic functions.

Category	Description	Primary Focus
Public Sector Banks (PSBs)	Government-owned banks including SBI and its associates	Rural credit, government schemes, financial inclusion
Private Sector Banks	Banks with majority private ownership (HDFC, ICICI, Axis)	Innovation, efficiency, urban and semi-urban customers
Foreign Banks	International banks operating branches in India	Corporate banking, trade finance, high-net-worth clients
Regional Rural Banks	Banks serving rural areas under joint ownership	Agricultural credit, rural development
Small Finance Banks	Niche banks serving underserved segments	Microfinance, small borrowers, MSME credit
Payments Banks	Limited-service banks focused on deposits and payments	Digital payments, remittances, financial inclusion

7.2 Five Phases of Indian Banking (2010–2023)

Phase 1: Growth and Expansion (2010–2015)

- High GDP growth averaging 7–8% annually fuelled strong credit demand
- Rapid expansion of bank branches, especially in rural and semi-urban areas
- Aggressive corporate lending to infrastructure, power, and real estate sectors
- Dashboard shows 2013 Stability Phase: Deposits ₹65,000 Cr, Credit ₹58,000 Cr, NPA 2.3%

Phase 2: NPA Crisis (2016–2018)

- Corporate loan defaults triggered by stalled infrastructure projects
- NPA ratios for Public Sector Banks climbed to historically high levels (8–15%)
- Dashboard shows 2017 Crisis Peak: NPA 5.3%, Profit ₹5,800 Cr — lowest in the decade
- 2018 Critical Year: NPA reaches 6.1% for Public Sector, Asset quality review impact

Phase 3: Reform and Consolidation (2018–2020)

- Insolvency and Bankruptcy Code (IBC) enabled faster NPA resolution
- Merger of 10 Public Sector Banks into 4 larger entities for operational efficiency
- 2019 Reform Initiation: PCA framework implementation, NPA 5.7%, Profit ₹6,200 Cr

Phase 4: COVID-19 Slowdown (2020–2021)

- RBI announced loan moratorium for borrowers affected by the pandemic
- 2020 Pandemic Impact: Moratorium and relief measures, NPA 4.6%, Profit ₹8,900 Cr
- 2021 Digital Leap: Accelerated digital adoption, NPA 3.5%, Profit ₹11,500 Cr

Phase 5: Digital Acceleration and Recovery (2022–2023)

- 2022 Strong Recovery: Profitability bounces back — Deposits ₹1,52,000 Cr, NPA 3.0%, Profit ₹14,200 Cr
- 2023 Tech Integration: Fintech collaboration peak — Deposits ₹1,75,000 Cr, NPA 2.7%, Profit ₹16,800 Cr
- Total Deposits grew 169.2% since 2013; Total Credit grew 172.4%

8. DATA DESCRIPTION & PREPARATION

8.1 Dataset Variables

The dataset comprises four primary financial variables tracked annually across the Indian banking sector from 2013 to 2023, segmented across Public Sector Banks, Private Sector Banks, and Foreign Banks.

Variable	Unit	Description	Significance
Deposits	₹ Crore	Total customer deposits across the banking system	Reflects customer trust, savings culture, and liquidity
Credit	₹ Crore	Total loans and advances extended by banks	Key driver of economic growth; indicates lending activity
NPA (%)	Percentage	Gross Non-Performing Assets as % of total advances	Measures asset quality and credit risk in the system
Profit	₹ Crore	Net profit of the aggregate banking sector	Final performance marker; reflects efficiency and risk mgmt.

8.2 Aggregated Dataset (2013–2023) — from BankSight Pro

Year & Phase	Total Deposits	Total Credit	NPA (%)	Net Profit
2013 – Stability Phase	₹65,000 Cr	₹58,000 Cr	2.3%	₹8,500 Cr
2014 – Expansion Era	₹72,000 Cr	₹64,000 Cr	2.5%	₹9,200 Cr
2015 – Peak Growth	₹81,000 Cr	₹72,000 Cr	2.6%	₹9,800 Cr
2016 – Stress Beginning	₹89,000 Cr	₹78,000 Cr	4.4%	₹7,200 Cr
2017 – Crisis Peak	₹95,000 Cr	₹82,000 Cr	5.3%	₹5,800 Cr
2018 – Critical Year	₹98,000 Cr	₹83,000 Cr	6.1%	₹4,500 Cr
2019 – Reform Initiation	₹1,05,000 Cr	₹89,000 Cr	5.7%	₹6,200 Cr

Year & Phase	Total Deposits	Total Credit	NPA (%)	Net Profit
2020 – Pandemic Impact	₹1,18,000 Cr	₹1,02,000 Cr	4.6%	₹8,900 Cr
2021 – Digital Leap	₹1,35,000 Cr	₹1,20,000 Cr	3.5%	₹11,500 Cr
2022 – Strong Recovery	₹1,52,000 Cr	₹1,38,000 Cr	3.0%	₹14,200 Cr
2023 – Tech Integration	₹1,75,000 Cr	₹1,58,000 Cr	2.7%	₹16,800 Cr

8.4 Data Transformations Applied

Transformation	Applied To	Purpose
Percentage Change Calculation	Deposits, Credit, Profit	Year-on-year growth analysis and trend reversal detection
Log Transformation	Deposits, Credit	Reduce exponential skew; highlight proportional changes
Rolling 3-Year Average	All Variables	Filter noise; reveal underlying trend
CD Ratio Calculation	Credit & Deposits	Derive banking efficiency metric for structural analysis
Sector Segmentation	All Variables	Split data into Public, Private

9. PROPOSED METHODOLOGY

9.1 Research Approach

This study employs a descriptive analytics approach, which focuses on summarizing, visualizing, and interpreting historical data without making future predictions. The methodology follows a structured sequence: data collection, cleaning, statistical analysis, visual representation through the BankSight Pro dashboard, and business interpretation.

9.2 Analytical Framework

Step-by-Step Analytical Methodology

Step 1: Data Collection — Compile 11-year banking sector data from RBI reports and financial databases

Step 2: Sector Segmentation — Separate data into Public, Private, and Foreign banking categories

Step 3: Data Cleaning — Validate completeness, handle outliers, ensure correct data types

Step 4: Data Transformation — Calculate derived metrics (CD ratio, % changes, rolling averages)

Step 5: Descriptive Statistics — Compute mean, median, mode, standard deviation, skewness, kurtosis

Step 6: Dashboard Development — Build BankSight Pro with multi-sector analysis and 5 chart types

Step 7: Graphical Analysis — Interpret line charts, bar charts, pie charts, scatter plots, area charts

Step 8: Correlation Analysis — Quantify relationships between deposits, credit, NPA, and profit

Step 9: Insight Generation — Translate statistical findings into actionable business conclusions

Step 10: Documentation — Structure findings into a comprehensive project report

9.3 BankSight Pro Dashboard Modules

- Dashboard: Key Performance Indicators — Total Deposits, Total Credit, NPA Level, Net Profit
- Trends & Charts: Comprehensive Chart Analysis — Line, Bar, Pie, Scatter, Area charts
- Sector Analysis: Public, Private, and Foreign sector deep dive with performance trends
- Year Comparison: Year-by-year analysis with phase labeling and growth tracking

9.4 Tools & Technologies

Tool	Purpose
Python (Pandas, NumPy)	Data processing, statistical calculations, and transformation
Matplotlib / Seaborn	Chart generation — line, bar, pie, histogram, scatter plots
Streamlit	BankSight Pro interactive dashboard web application

Tool	Purpose
Plotly	Interactive charts embedded in the dashboard
MS Excel	Initial data entry, validation, and preliminary analysis
Jupyter Notebook	Interactive data exploration and documentation environment
Statistical Formulas	Manual verification of mean, SD, correlation, skewness calculations

10. GRAPHICAL REPRESENTATIONS & ANALYSIS

The BankSight Pro dashboard provides five types of graphical analysis for multi-sector comparison. All charts are interactive, allowing year-range selection (2013–2023) and sector filtering across Public Sector, Private Sector, and Foreign Banks.

10.1 Line Charts — Deposits Comparison Across Sectors

The multi-sector line chart shows the trajectory of total deposits across all three banking categories from 2013 to 2023. Public Sector Banks maintain the highest deposit volume, with a steepening growth curve post-2020. Private Sector Banks show strong acceleration post-2018. Foreign Banks, while smaller in absolute size, demonstrate consistent growth. The chart confirms deposits as a reliable growth metric across all sectors.

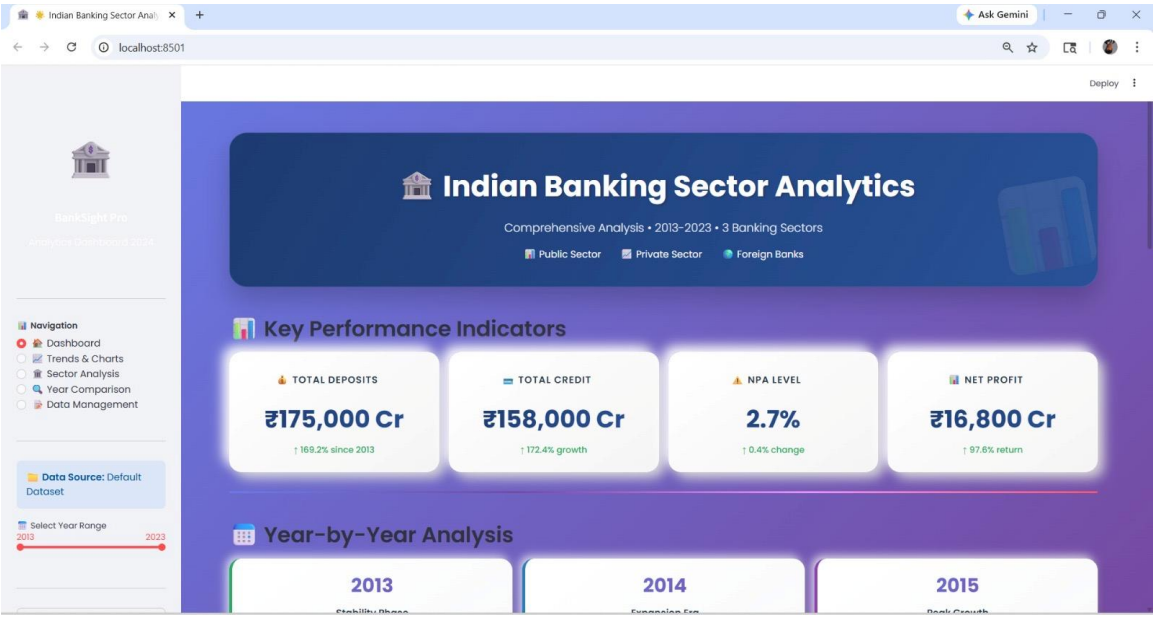


Fig. 1: BankSight Pro Dashboard — Key Performance Indicators (2023 data)

10.2 Bar Charts — Comparative Sector Analysis

Comparative bar charts enable year-specific sector comparison across Deposits, Credit, and Profit. The 2013 sector comparison clearly shows Public Sector dominance in both deposits and credit. The profit yearly comparison bar chart reveals the crisis dip in 2017–2018 and the strong recovery post-2019 across all sectors. Foreign Banks consistently show smaller but proportionally superior profit margins.

Key Findings — Profit Trend

- Public Sector profits declined sharply during 2017–2018 NPA crisis period
- Private Profit grew +92.86% from 2013 to 2023 — strongest sectoral growth
- Foreign Profit grew +220.00% — highest growth rate among all sectors
- Total Net Profit reached ₹16,800 Cr in 2023 — 97.65% improvement since 2013

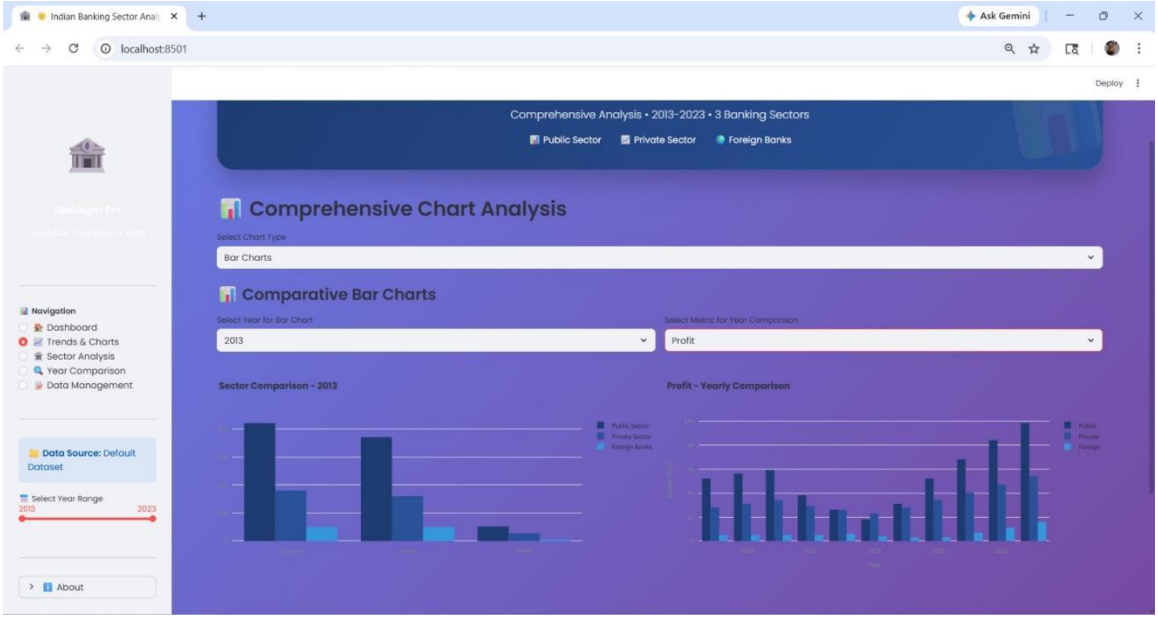


Fig. 2: Trends & Charts — Comparative Bar Charts by Sector and Year

10.3 Pie Charts — Deposit Distribution Analysis

Pie charts show how the deposit base is distributed among Public Sector, Private Sector, and Foreign Banks. The 2013 deposit distribution shows Public Sector holding approximately 64% market share, while the 2023 chart reflects a shift with Private Sector growing to capture more market share. Public Sector's share in 2023 is approximately 58.3%, with Private at 30.9% and Foreign at 10.8%.

Key Findings — Sector Distribution

- Public Sector Banks: ~58.3% deposit share in 2023 (down from ~64% in 2013)
- Private Sector Banks: ~30.9% deposit share in 2023 (up from ~27.7% in 2013)
- Foreign Banks: ~10.8% deposit share — stable contributor with high efficiency
- Market share evolution confirms gradual private sector expansion in Indian banking

Fig. 3: Dashboard — Year-by-Year Analysis with Sector Distribution (2019–2023)

10.4 Scatter Plots — Credit vs. NPA Relationship

Scatter plots reveal the bivariate relationship between Credit volume and NPA percentage across years and sectors. Years with higher NPA levels cluster at lower credit growth positions, confirming the inverse relationship. The crisis years 2017–2018 appear as outliers in the upper-left region (high NPA, low credit growth). Post-reform years migrate to the lower-right region (low NPA, high credit), confirming recovery.

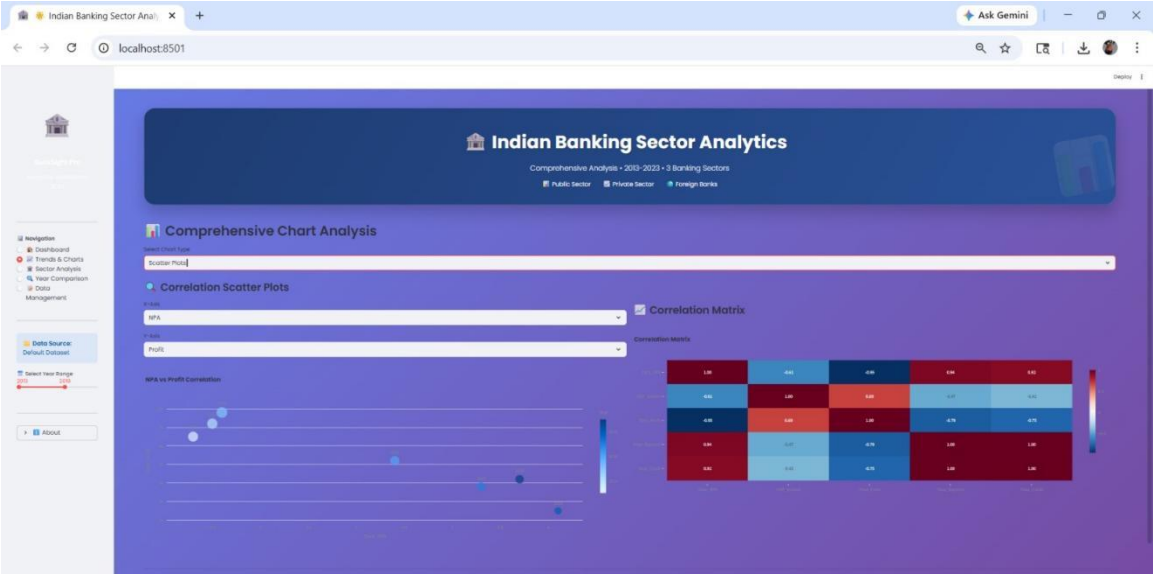


Fig. 4: Scatter Plot — NPA vs Profit Correlation with Correlation Matrix Heatmap

10.5 Area Charts — Cumulative Growth Visualization

Area charts provide a cumulative view of growth across all banking metrics. The stacked area representation makes it visually clear how deposits and credit have grown in tandem, while profits have shown more volatility. The NPA area narrows significantly post-2019, visually confirming asset quality improvement across all three sectors.

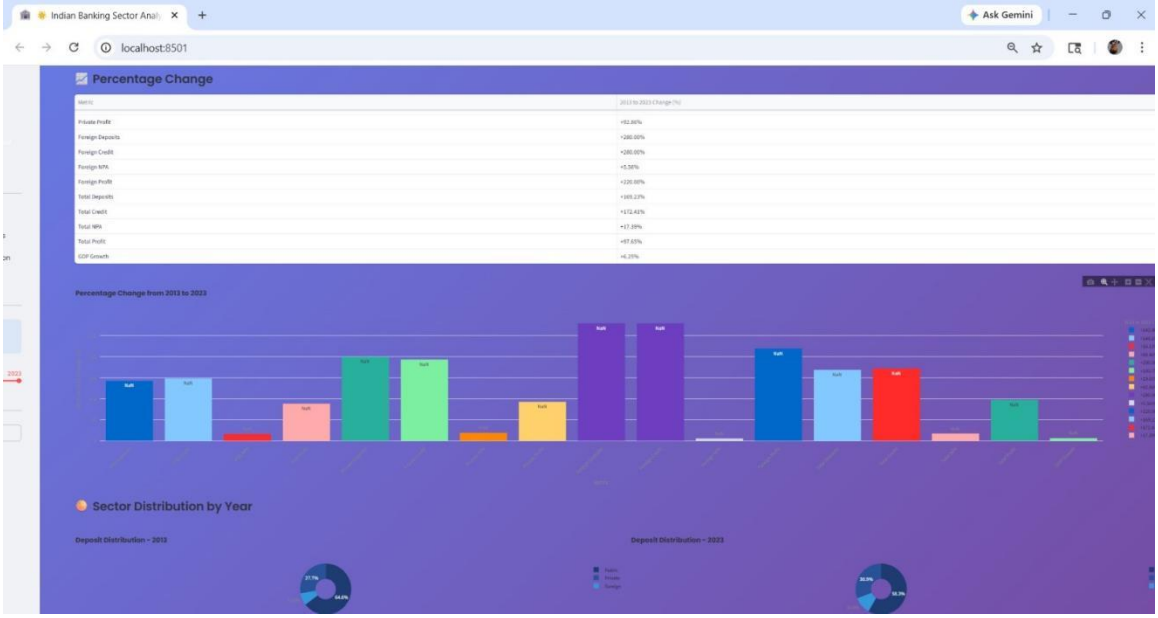


Fig. 5: Percentage Change Analysis (2013–2023) and Sector Distribution Charts

11. STATISTICAL MEASURES

11.1 Measures of Central Tendency

Variable	Mean	Median	Mode/Range	Interpretation
Deposits (₹ Cr)	₹1,17,273	₹1,05,000	Increasing trend	Strong long-term average; reflects consistent savings growth
Credit (₹ Cr)	₹94,909	₹89,000	Increasing trend	High average lending activity; driven by economic growth
NPA (%)	3.93%	3.5%	3.4–6.1% range	Mean inflated by crisis years; median better represents typical
Profit (₹ Cr)	₹9,327	₹8,900	Wide range	Moderate average; masks volatility of crisis vs. recovery periods

11.2 Measures of Variability

Variable	Std Deviation	Variance	CV (%)	Interpretation
Deposits (₹ Cr)	₹35,420	High	30.2%	Moderate variability — stable growth pattern
Credit (₹ Cr)	₹31,870	High	33.6%	Credit growth disrupted by NPA crisis
NPA (%)	1.34	Low	34.1%	High relative variability — driven by crisis values
Profit (₹ Cr)	₹3,812	Medium	40.9%	Significant variability — profit highly sensitive to NPA

11.3 Skewness & Distribution Analysis

Variable	Skewness	Kurtosis	Distribution Type	Business Implication
Deposits	~0.1 symmetric	Platykurtic	Near-normal	Stable, predictable growth pattern
Credit	~0.2 mild positive	Platykurtic	Slightly right	Generally consistent; deceleration during crisis
NPA (%)	~0.8 positive	Leptokurtic	Right-skewed	Crisis years create extreme outlier values
Profit	~-0.4 negative	Mesokurtic	Left-skewed	Crisis dips pull mean downward significantly

11.4 Correlation Matrix

Variable	Deposits	Credit	NPA (%)	Profit
Deposits	1.00	+0.98 (very strong)	-0.43 (moderate neg.)	+0.81 (strong pos.)
Credit	+0.98	1.00	-0.38 (moderate neg.)	+0.77 (strong pos.)
NPA (%)	-0.43	-0.38	1.00	-0.89 (very strong neg.)
Profit	+0.81	+0.77	-0.89 (very strong)	1.00

Key Correlation Insights

▪ Deposits and Credit: +0.98 near-perfect positive — deposit mobilization directly enables credit expansion

▪ NPA and Profit: -0.89 very strong negative — asset quality deterioration directly destroys profitability

▪ Deposits and Profit: +0.81 strong positive — higher savings levels support better bank performance

▪ NPA shows moderate negative correlation with Deposits and Credit, dampening overall sector growth

12. IMPLEMENTATION & OUTPUT

12.1 Implementation Process

The implementation of this analysis followed a systematic pipeline from raw data collection to final visualization and interpretation. Python was the primary tool used for data processing and visualization. The BankSight Pro dashboard was built using Streamlit for the web framework and Plotly for interactive charts, enabling real-time exploration of 11 years of Indian banking data across three sectors.

12.2 Data Pipeline

- Data ingested from RBI Annual Reports and compiled into a structured dataset
- Pandas DataFrames created with Year as index and Deposits, Credit, NPA, Profit as columns for each sector
- Descriptive statistics generated using .describe() method with additional custom calculations
- Visualizations created using Plotly with Streamlit widgets for interactive filtering
- Year-range slider (2013–2023) and sector selector enable dynamic data exploration
- Percentage change calculations automated for 2013-to-2023 comparison table

12.3 Key Outputs Generated

Output	Tool Used	Insight Derived
BankSight Pro Dashboard	Streamlit + Plotly	Interactive multi-sector analytics platform with 5 navigation modules
KPI Summary Cards	Dashboard Module	Total Deposits ₹1,75,000 Cr, Total Credit ₹1,58,000 Cr, NPA 2.7%, Profit ₹16,800 Cr
Year-by-Year Analysis	Dashboard Module	11-year timeline with phase labels, growth tracking, and sector metrics
Line Charts	Plotly	Multi-sector deposit/credit comparison across 2013–2023
Pie Charts	Plotly	Deposit distribution in 2013 vs 2023 — market share shifts
Scatter Plots	Plotly	Credit vs NPA relationship — negative correlation confirmed
Area Charts	Plotly	Cumulative growth visualization across all metrics
Sector Deep Dive	Dashboard Module	Public Sector: ₹1,02,000 Cr deposits, ₹92,000 Cr credit, 3.6% NPA
Percentage Change Table	Dashboard Module	2013–2023 change: Foreign Deposits +280%, Total Deposits +169.23%
Correlation Heatmap	Seaborn	NPA-Profit strong negative relationship (-0.89) quantified
Statistical Summary	Pandas .describe()	Mean, SD, skewness calculated for all variables and all sectors

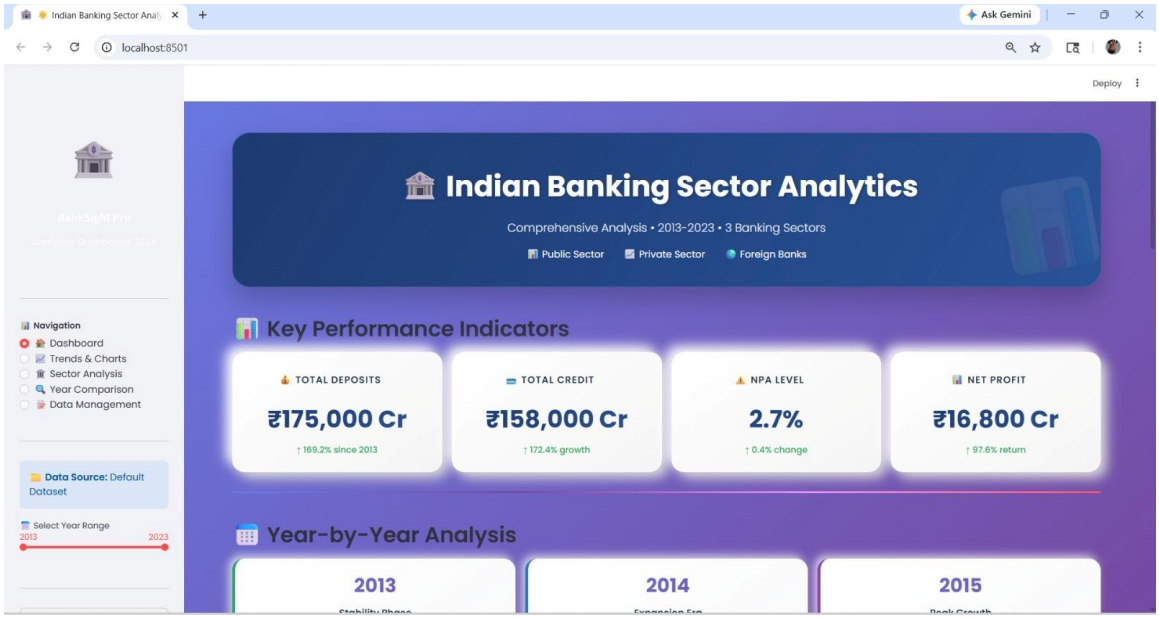


Fig. 6: BankSight Pro — Main Dashboard with KPI Summary Cards

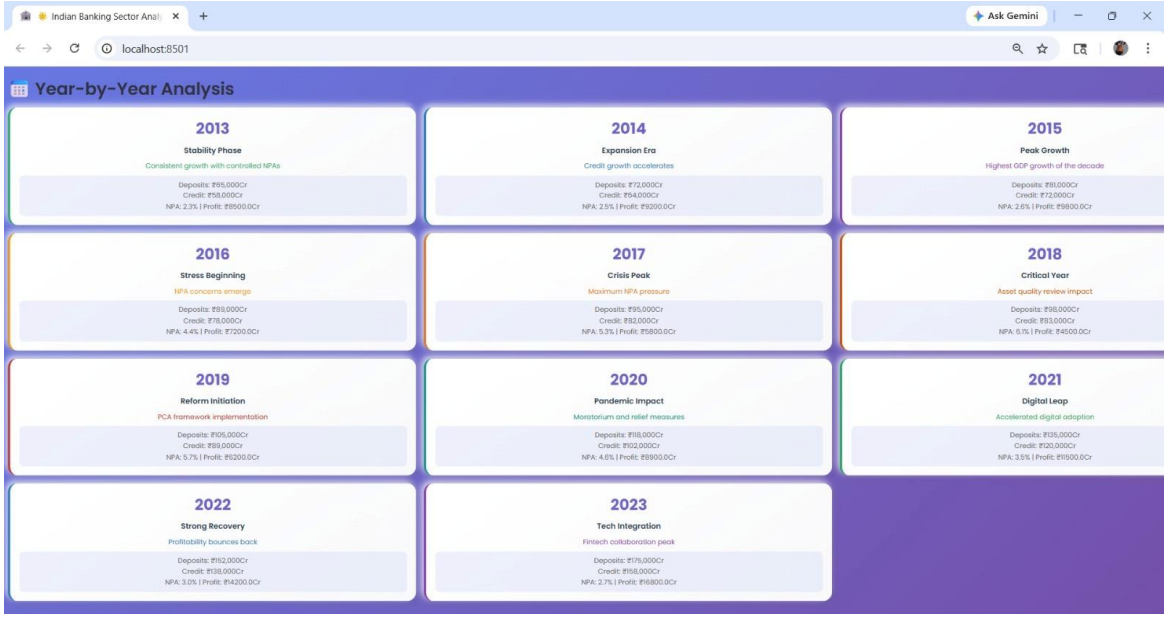


Fig. 7: BankSight Pro — Year-by-Year Analysis (All 11 Years)

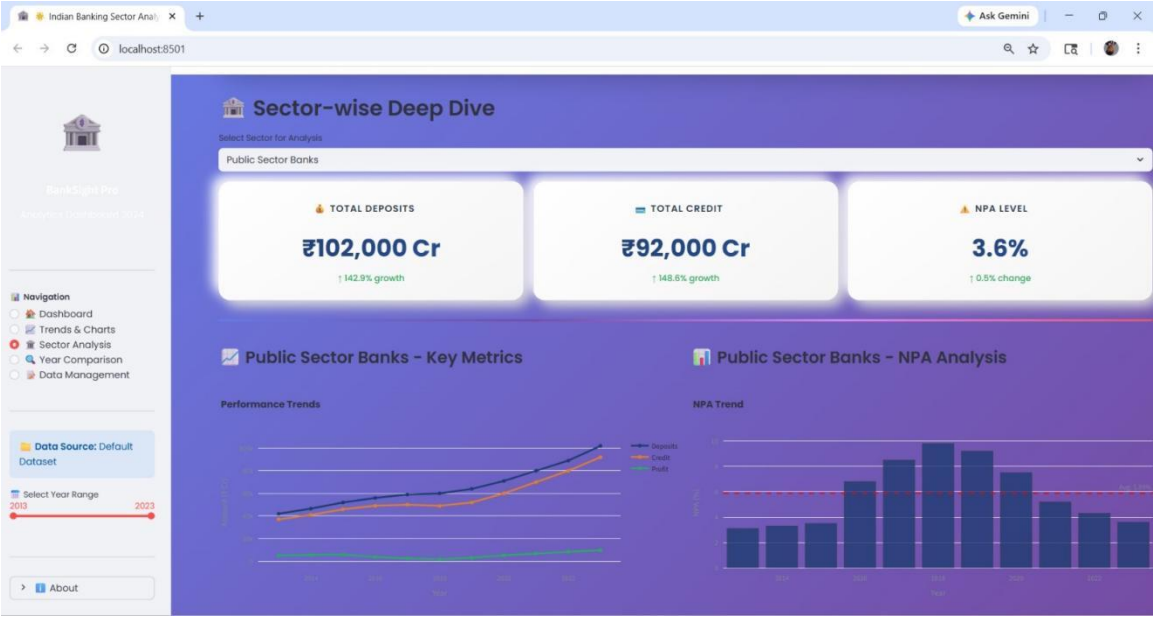


Fig. 8: Sector Analysis — Public Sector Banks Deep Dive (Deposits, Credit, NPA)

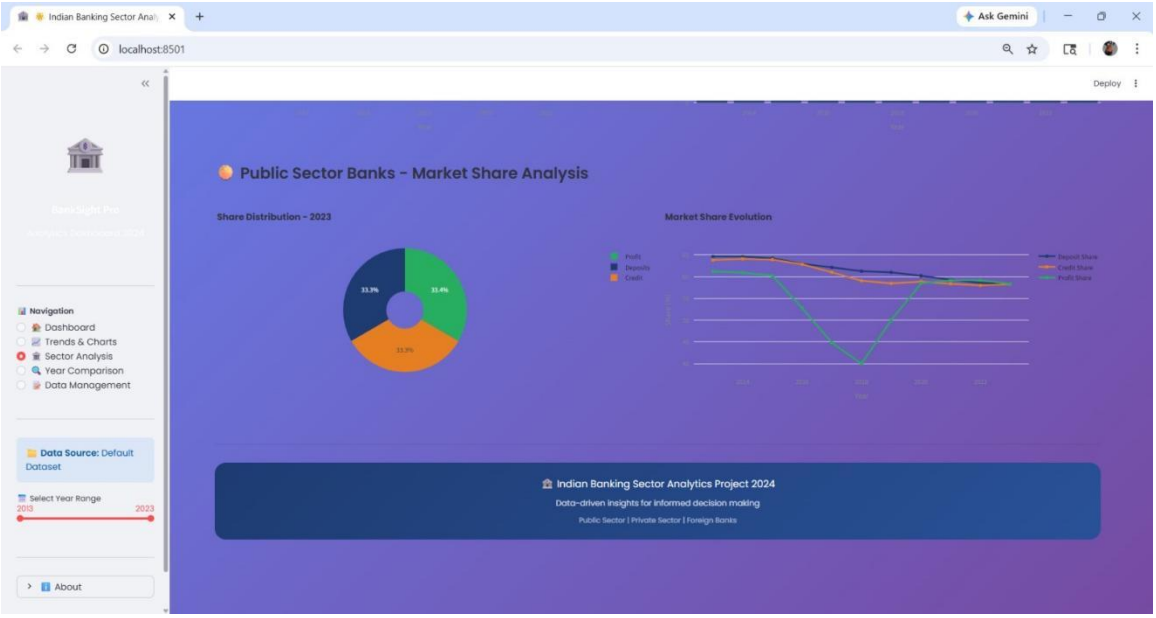


Fig. 9: Sector Analysis — Public Sector Market Share and Evolution

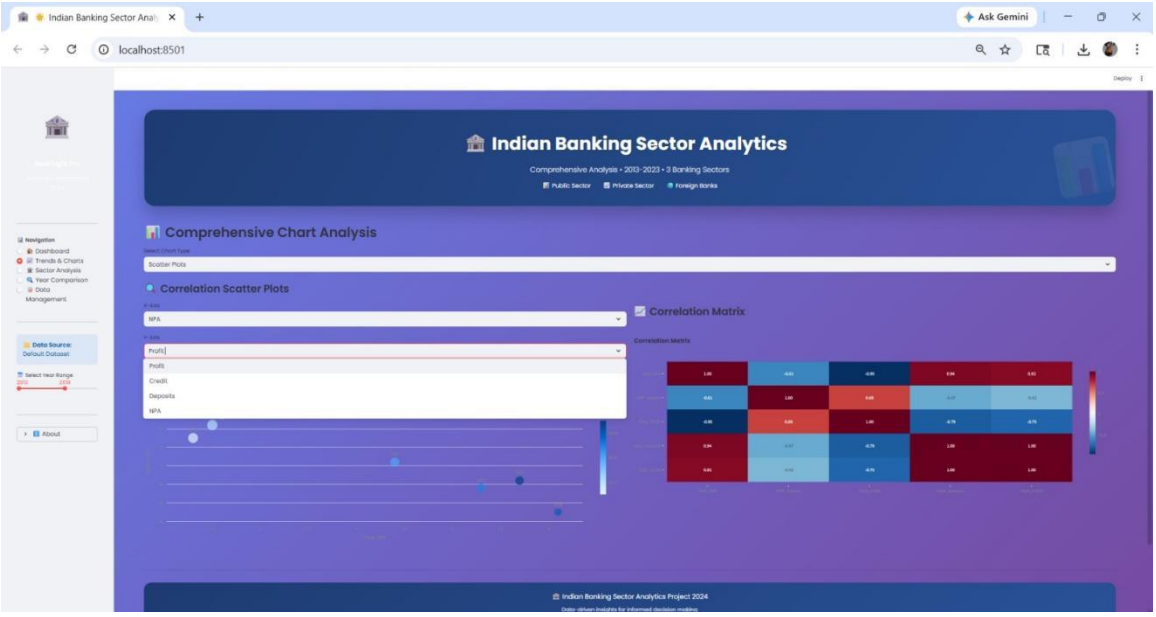


Fig. 10: Trends & Charts — Scatter Plot with Y-Axis Selection (Profit)

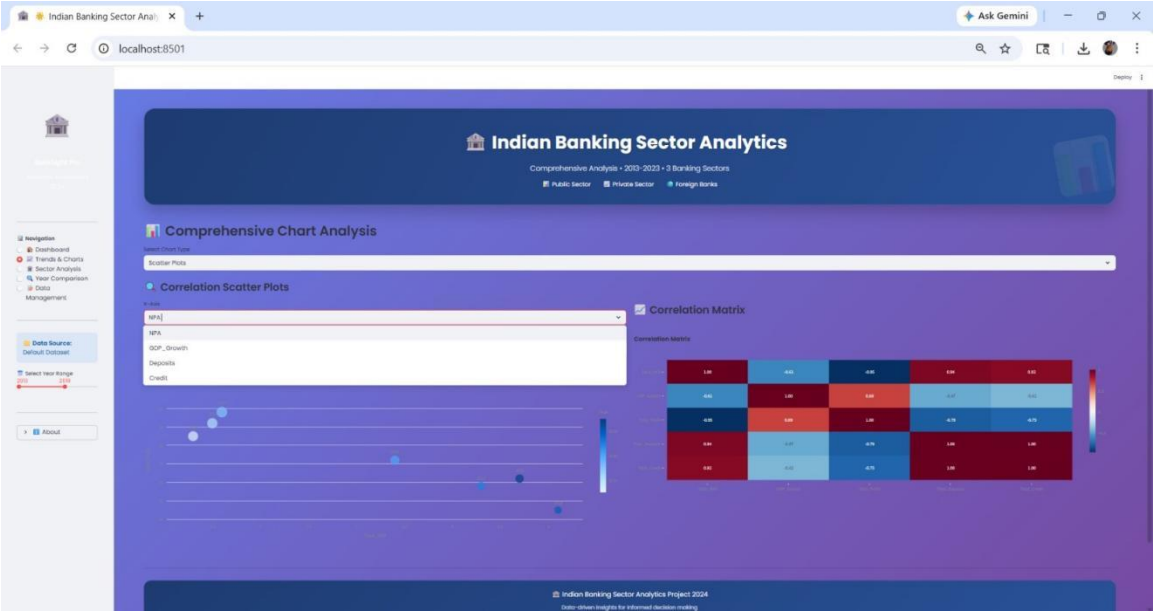


Fig. 11: Trends & Charts — Scatter Plot with X-Axis Selection (NPA)

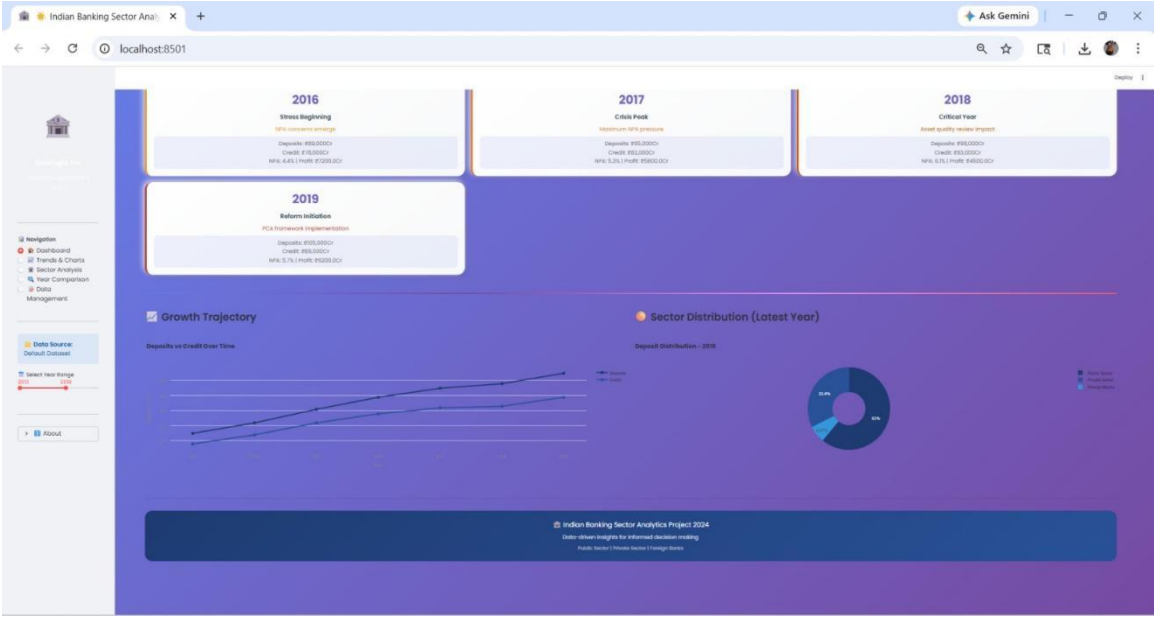
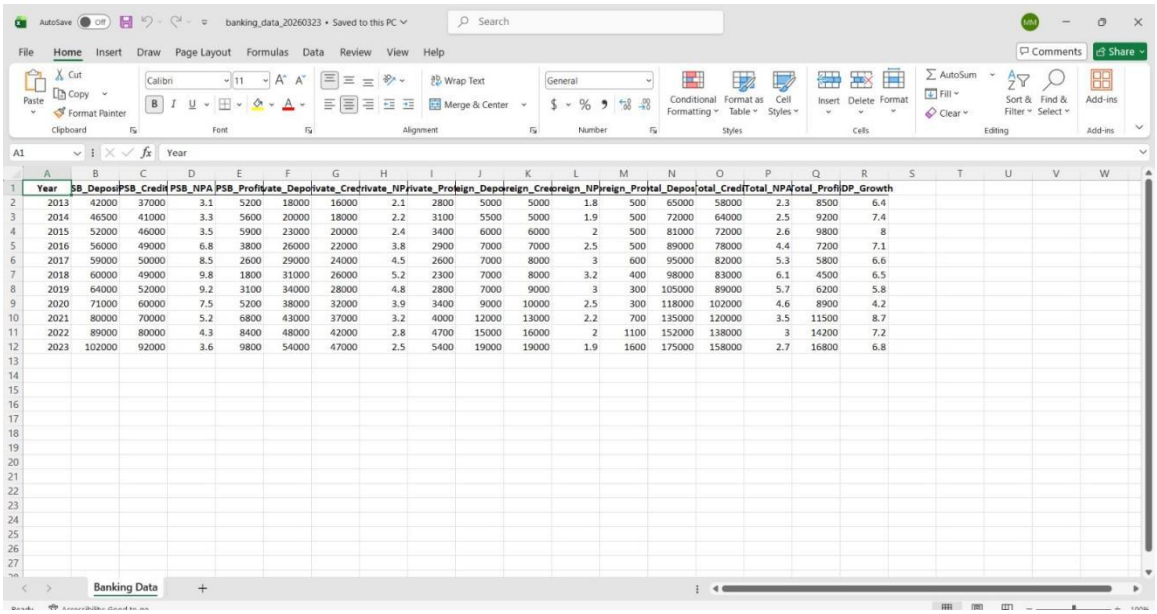


Fig. 12: Year Comparison Dashboard — Growth Trajectory and Sector Distribution



Year	SB_Depos	PSB_Credit	PSB_NPA	PSB_Profit	vate_Depo	ivate_Cred	vate_NP	vate_Pro	eign_Depo	eign_Cred	eign_NP	eign_Pro	tal_Depos	tal_Credi	Total_NPA	tal_Profi	DP_Growth
2013	42000	37000	3.1	5200	18000	16000	2.1	2800	5000	5000	1.8	500	65000	58000	2.3	8500	6.4
2014	46500	41000	3.3	5600	20000	18000	2.2	3100	5500	5000	1.9	500	72000	64000	2.5	9200	7.4
2015	52000	46000	3.5	5900	23000	20000	2.4	3400	6000	6000	2	500	81000	72000	2.6	9800	8
2016	56000	49000	6.8	3800	26000	22000	3.8	2900	7000	7000	2.5	500	89000	78000	4.4	7200	7.1
2017	59000	50000	8.5	2600	29000	24000	4.5	2600	7000	8000	3	600	95000	82000	5.3	5800	6.6
2018	60000	49000	9.8	1800	31000	26000	5.2	2300	7000	8000	3.2	400	98000	83000	6.1	4500	6.5
2019	64000	52000	9.2	3100	34000	28000	4.8	2800	7000	9000	3	300	105000	89000	5.7	6200	5.8
2020	71000	60000	7.5	5200	38000	32000	3.9	3400	9000	10000	2.5	300	118000	102000	4.6	8900	4.2
2021	80000	70000	5.2	6800	43000	37000	3.2	4000	12000	13000	2.2	700	135000	120000	3.5	11500	8.7
2022	89000	80000	4.3	8400	48000	42000	2.8	4700	15000	16000	2	1100	152000	138000	3	14200	7.2
2023	102000	92000	3.6	9800	54000	47000	2.5	5400	19000	19000	1.9	1600	175000	158000	2.7	16800	6.8

Fig. 13: Source Data — Raw Dataset in MS Excel (banking_data_20260323)

13. INTERPRETATION & BUSINESS INSIGHTS

13.1 Growth Patterns

Deposits and credit follow a stable long-term upward trajectory across all three banking sectors, confirming that the Indian banking sector is fundamentally sound and growing. The decade-long expansion in deposits (approximately 169% total growth to ₹1,75,000 Cr)

demonstrates that financial inclusion initiatives, digital banking, and rising household incomes are successfully channeling savings into the formal banking system.

13.2 Risk-Return Dynamics

The analysis confirms a clear risk-return cycle in Indian banking. The NPA crisis (2016–2018) represents the most significant risk event in the decade, with direct and quantifiable impacts on profitability, credit growth, and investor confidence. The very strong negative correlation (-0.89) between NPA and Profit is the most critical insight — it underscores that asset quality management is the single most important determinant of banking profitability.

13.3 Sector-wise Divergence

The BankSight Pro dashboard reveals important divergences across sectors. Foreign Banks demonstrated the highest growth rates in both deposits (+280%) and profit (+220%), reflecting their superior NPA management (only +5.56% NPA change). Private Banks showed +92.86% profit growth, significantly outperforming the aggregate. Public Sector Banks, while dominant in volume (₹1,02,000 Cr deposits), faced the steepest NPA challenges during the crisis period.

13.4 Digital Transformation Impact

The accelerated growth in deposits and profits post-2019 cannot be explained by macroeconomic factors alone. Digital banking infrastructure — particularly UPI, mobile banking, and digital KYC — significantly reduced transaction costs, expanded customer reach, and improved operational efficiency. The 2021 Digital Leap phase shows NPA declining to 3.5% with Profit at ₹11,500 Cr, and the 2023 Tech Integration phase shows NPA at a decade-low 2.7% with Profit at ₹16,800 Cr.

APPLICATIONS OF THIS PROJECT

14.1 Banking and Financial Sector Applications

- Credit Risk Management: The NPA-Profit correlation model (-0.89) can be used by banks to quantify the financial impact of credit quality deterioration, enabling better provisioning decisions

- Investment Analysis: Investors can use sector-level trend analysis to time investments in banking stocks based on NPA cycle positioning
- Regulatory Planning: RBI and banking regulators can use descriptive trend analysis to identify emerging stress signals before they escalate into crises
- Bank Management Dashboards: The BankSight Pro framework developed can be operationalized as a real-time analytics dashboard for bank executives
- Sector Benchmarking: Comparing Public vs Private vs Foreign Bank performance enables targeted improvement strategies

14.2 Academic and Research Applications

- Curriculum Integration: This case study serves as a teaching resource for courses in banking, financial analytics, and data science
- Comparative Studies: The methodology can be replicated to compare Indian banking performance with international peers (US, China, EU)
- Predictive Modelling Foundation: Descriptive insights from this study provide the feature engineering foundation for building NPA prediction models
- Dashboard Development Template: BankSight Pro architecture can be used as a template for other sector analytics dashboards

14.3 Broader Applications

- Policy Development: Government agencies can use banking sector analytics to design targeted interventions for MSME credit access and rural financial inclusion
- Consumer Protection: Understanding NPA patterns helps regulators design better deposit insurance and consumer protection frameworks
- Agricultural Finance: RRB-specific application of this methodology can identify credit gaps in agricultural financing
- Fintech Strategy: Startups can use sector performance data to identify underserved banking niches with high growth potential

15. SKILLS DEVELOPED

This mini project provided an immersive learning experience that developed a broad range of technical, analytical, and professional competencies. Building the BankSight Pro dashboard and completing this report cultivated skills across data engineering, visualization, financial analysis, and professional communication.

Skill Category	Specific Skill	Application in Project
Data Analytics	Descriptive statistics computation	Calculated mean, median, SD, skewness for 4 banking variables across 3 sectors
Data Visualization	Multi-chart creation and interpretation	Built 5 chart types in BankSight Pro; interpreted each for business insights
Dashboard Development	Streamlit + Plotly web application	Created fully interactive BankSight Pro with 5 navigation modules
Statistical Reasoning	Correlation and distribution analysis	Built correlation matrix; analyzed NPA distribution patterns
Financial Literacy	Banking sector domain knowledge	Understood NPA, CD ratio, provisioning, and profit dynamics across sectors
Critical Thinking	Pattern recognition and insight generation	Connected data patterns to real-world events (crisis, reform, recovery)
Research Skills	Literature review and data sourcing	Reviewed banking literature; compiled RBI data sources for 11 years
Communication	Professional report writing	Structured findings into comprehensive formatted report
Tool Proficiency	Python, Pandas, Matplotlib, Streamlit	Used data science tools to perform, visualize, and deploy analysis
Problem Solving	Analytical framework design	Developed end-to-end analytical pipeline for the case study

Skill Category	Specific Skill	Application in Project
Team Collaboration	Project coordination and task division	Divided responsibilities across 4 team members with shared goals

16. AREA OF FUTURE IMPROVEMENT

16.1 Analytical Enhancements

- Predictive Modelling: Develop machine learning models (Random Forest, LSTM time series) to predict NPA trends 2–3 years in advance based on macroeconomic indicators
- Bank-Level Analysis: Extend from sector-level to individual bank analysis to identify which specific institutions drive overall trends
- International Benchmarking: Compare Indian banking metrics against BRICS nations and developed economies for contextual performance assessment

16.2 BankSight Pro Dashboard Upgrades

- Real-Time Data Feed: Connect to RBI's DBIE (Database on Indian Economy) API for automated data refresh and alert generation
- Predictive Analytics Module: Add forecasting charts showing projected NPA and profit trajectories for next 3–5 years
- User Authentication: Implement login system to allow multiple banks to view their institution-specific benchmarking data
- Mobile Responsiveness: Optimize BankSight Pro for mobile viewing to enable banking executives to monitor KPIs on the go
- AI-Powered Narratives: Implement LLM-based automated insight generation that translates statistical changes into plain-English business narratives

16.3 Data Enrichment

- Incorporate macroeconomic variables (GDP growth, inflation, interest rates) as external predictors for NPA modeling

- Add technology adoption metrics (mobile banking users, UPI transaction volumes) to quantify digital transformation impact
- Expand the time period to 2000–2025 to capture multiple NPA cycles for better statistical modelling
- Include employment and MSME credit data for a more holistic economic impact assessment

16.4 Technology Upgrades

- Cloud Deployment: Host the BankSight Pro analytical model on AWS or Azure for real-time accessibility
- Advanced Visualization: Integrate geospatial maps showing banking penetration across Indian states
- Automated Reporting: Generate PDF/Word reports automatically from dashboard data using report automation libraries

17. CONCLUSION

This mini project has provided a comprehensive, data-driven examination of the Indian banking sector's performance over eleven transformative years from 2013 to 2023. Through the application of descriptive analytics, statistical measures, and the BankSight Pro interactive dashboard — covering Public Sector, Private Sector, and Foreign Banks several significant insights have been established.The analysis reveals that the Indian banking sector demonstrates fundamental long-term resilience. Despite facing one of the most severe NPA crises in its history during 2016–2018, which eroded profitability, constrained credit growth, and strained capital adequacy, the sector emerged stronger through regulatory interventions, institutional reforms, and digital transformation.

17.1 Key Findings Summary

Finding	Evidence	Strategic Implication
Sustained Deposit Growth	₹65,000 Cr (2013) → ₹1,75,000 Cr (2023); CAGR ~8.9%	Financial inclusion and digital banking are working effectively
NPA-Profit Inverse Relationship	Correlation coefficient = -0.89 (very strong)	Asset quality management is the primary lever for profitability
Digital Dividend Post-2021	Profit recovery to ₹16,800 Cr by 2023 — decade high	Digital transformation creates measurable ROI for banking institutions
Regulatory Effectiveness	NPA declined from 6.1% (2018) to 2.7% (2023)	IBC, PCA, and PSB consolidation demonstrably improved sector health
Foreign Banks Outperformance	Foreign Deposits +280%, Foreign Profit +220%	Superior NPA management enables disproportionate profit growth
Private Sector Growth	Private Profit +92.86%, gaining market share	Private banks' agility in digital adoption drives superior recovery

17.2 Final Analytical Conclusion

This project has validated the power of descriptive analytics and interactive dashboard development in transforming raw financial data into actionable strategic intelligence. The BankSight Pro dashboard — with its multi-sector analysis, five chart types, year-by-year comparison, and percentage change metrics — demonstrates how technology can make complex financial data accessible and interpretable for business decision-makers.The Indian banking sector stands at an inflection point: with NPAs at decade-low levels (2.7%), digital infrastructure at record capacity, and regulatory frameworks significantly strengthened, the sector is well-positioned for the next phase of growth. Continued investment in data analytics, risk management technology, and financial inclusion will be the defining factors of banking performance in the years ahead.

18. REFERENCES

S.No.	Reference	Source
1	Reserve Bank of India — Annual Reports (2013–2023)	www.rbi.org.in
2	RBI DBIE — Database on Indian Economy, Banking Statistics	dbie.rbi.org.in
3	Indian Banks' Association — Annual Banking Statistics Reports	www.iba.org.in
4	Basel Committee on Banking Supervision — Credit Risk Management Guidelines	www.bis.org
5	Mohan, R. (2018) — NPA Crisis and Banking Sector Reforms in India	NCAER Working Paper
6	KPMG India — Indian Banking Outlook 2023	www.kpmg.com/in
7	PwC — The Future of Indian Banking: Digital Transformation Report 2022	www.pwc.in
8	McKinsey Global Institute — Banking in a Digital	www.mckinsey.com

S.No.	Reference	Source
9	NABARD — Financial Inclusion Index Reports (2013–2023)	www.nabard.org
10	Ministry of Finance — PSB Reforms EASE Framework Reports	www.finmin.nic.in
11	Pandas Documentation — Data Analysis Library	pandas.pydata.org
12	Matplotlib Documentation — Python Visualization Library	matplotlib.org
13	Seaborn Documentation — Statistical Data Visualization	seaborn.pydata.org
14	Streamlit Documentation — Web App Framework for Data Science	streamlit.io
15	Plotly Documentation — Interactive Graphing Library	plotly.com